📄 Problem Statement Document

VLSI: Simulation of Finite State Machine (FSM) using ModelSim

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1. 📌 Title

Simulation-Based Design and Verification of Finite State Machine (FSM) using ModelSim

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2. ❗ Problem Statement

Finite State Machines (FSMs) are a fundamental concept in VLSI design, widely used in control units, communication systems, and digital circuits. Designing FSMs directly on hardware can be complex, error-prone, and difficult to debug, especially when dealing with multiple states, transitions, and timing constraints.

Additionally, without proper simulation tools, verifying the correctness of state transitions and output behavior becomes challenging. Errors in FSM design can lead to incorrect system functionality and costly redesigns.

Therefore, there is a need for a simulation-based approach to design, implement, and verify FSM behavior. Using simulation tools allows designers to validate logic, observe state transitions, and ensure correct operation before hardware implementation.

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3. 🎯 Proposed Solution

The proposed system is a **simulation-based FSM design and verification project** using industry-standard tools.

Simulation Tool:

- ModelSim

Key Components (Simulated):

- FSM Design (Moore or Mealy machine)
- HDL Code (Verilog / VHDL)
- Testbench for simulation
- Waveform viewer for analysis

Working:

- FSM is designed using HDL (Verilog/VHDL)
- Inputs are applied through a testbench
- State transitions occur based on input conditions
- Outputs are generated depending on FSM type
- Simulation waveforms are analyzed to verify correctness

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4. 🎯 Objectives

- To design a Finite State Machine using HDL
- To simulate FSM behavior using ModelSim
- To verify state transitions and outputs
- To understand Moore and Mealy machine concepts
- To debug digital designs using simulation tools

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5. 🌍 Scope of the Problem

FSMs are widely used in:

- Digital system design
- Control units in processors
- Communication protocols
- Embedded and real-time systems

This simulation-based approach helps in:

- Academic learning
- Prototype validation
- VLSI design verification

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6. ⚠️ Constraints

- Simulation does not include physical hardware limitations
- Timing behavior may differ in real implementations
- Requires understanding of HDL coding

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7. 📌 Expected Outcome

- Correct FSM implementation in HDL
- Verified state transitions using simulation
- Waveform outputs showing system behavior
- Strong understanding of VLSI design concepts

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8. 📎 Submission Instructions

Option 1: Google Drive

- Upload:
  - Problem Statement PDF
  - HDL code (Verilog/VHDL)
  - Testbench file

- Simulation waveform screenshots
- Share via Google Drive
- Set access → **Anyone with the link**

Option 2: Google Colab

- Use Google Colab
- Include:
  - FSM explanation (Markdown)
  - Code snippets
  - Output waveform images
- Share → **Anyone with the link**
